Diabetic Emergency - Adult

AM-02 | Adult Medical

Quick flow

01 Check Glucose + ABCs [EMT / AEMT / Paramedic]

Altered mental status • weakness • seizure • diaphoresis • vomiting • suspected diabetes

02 Glucose <70 mg/dL? [EMT / AEMT / Paramedic]

Treat when symptomatic or when symptoms cannot be assessed reliably because of altered mental status

03 Safe Swallow? [EMT / AEMT / Paramedic]

Yes: oral glucose 15-25 g • No: AEMT/Paramedic D10 IV/IO or glucagon IM

04 Titrate D10 [AEMT / Paramedic]

Up to 250 mL IV/IO • stop when mental status improves and glucose ≥70 • maximum 25 g

05 Reassess [EMT / AEMT / Paramedic]

Parenteral: glucose + neuro exam at 5 min • oral: repeat glucose within 10-15 min

06 Hyperglycemia / DKA Features? [EMT / AEMT / Paramedic]

Glucose ≥250 or Kussmaul breathing, vomiting, abdominal pain, dehydration, infection, shock, or SGLT2 use

07 Hypotension / Poor Perfusion? [AEMT / Paramedic]

Isotonic fluid 500 mL IV/IO aliquots • 250 mL if volume-overload risk • maximum 2 L

08 Transport / Refusal Screen [EMT / AEMT / Paramedic]

Transport high-risk medications, recurrence, seizure, pregnancy, overdose, uncertain cause, DKA/HHS, or persistent deficit

Overview

- Treat symptomatic hypoglycemia promptly while protecting the airway and identifying stroke, seizure, overdose, trauma, infection, or another concurrent emergency.
- Use titrated D10 rather than a fixed hypertonic dextrose bolus. The treatment target is clinical improvement and glucose at least 70 mg/dL - not routine hyperglycemia.
- Marked hyperglycemia requires assessment for DKA, HHS, infection, dehydration, and shock. Claiborne EMS does not administer insulin under this protocol.

Provider-Level Actions

Correct hypoglycemia safely, identify high-risk recurrence, and provide supportive care for hyperglycemic emergencies.

EMT

- Assess airway, breathing, circulation, mental status, and neurologic findings; obtain blood glucose early and repeat it whenever the patient's condition changes.
- When glucose is below 70 mg/dL and the patient is symptomatic, awake, follows commands, swallows reliably, and protects the airway, give oral glucose 15-25 g according to the packaged dose.
- Do not give oral glucose, food, fluid, or medication when swallowing or airway protection is impaired. Request ALS and provide aspiration protection and ventilatory support as indicated.
- Identify diabetes medications, last dose, last meal, treatment before EMS arrival, recurrent episodes, and high-risk transport features.

AEMT

- Perform all EMT actions and establish IV access when parenteral treatment is required; use IO access when urgently needed and IV access cannot be obtained within the Universal Patient Care attempt limit.
- When oral treatment is unsafe or ineffective, give D10 up to 250 mL IV/IO, titrated to improving mental status and glucose at least 70 mg/dL, maximum 25 g.
- When oral treatment is unsafe and vascular access cannot be obtained promptly, give glucagon 1 mg IM and protect against aspiration because vomiting may occur.
- For hypotension or poor perfusion, give isotonic crystalloid in 500 mL aliquots with reassessment, maximum 2 liters; use 250 mL aliquots when volume-overload risk is present.

Paramedic

- Perform all EMT and AEMT actions and evaluate persistent symptoms for stroke, seizure, overdose, trauma, sepsis, dysrhythmia, hypoxia, or another time-sensitive condition.
- Obtain continuous cardiac monitoring and a 12-lead ECG when dysrhythmia, ischemia, severe electrolyte disturbance, significant hyperglycemic illness, or another cardiac indication is suspected.
- Manage airway failure, seizure, shock, and associated emergencies under the corresponding linked protocol.
- Claiborne EMS does not administer insulin or independently change an insulin-pump dose under AM-02.

Indications

- Symptoms or altered mental status with suspected hypoglycemia or hyperglycemia.
- Known diabetes with abnormal glucose, medication complication, insulin-pump problem, or inability to maintain oral intake.
- Suspected DKA, HHS, euglycemic DKA, or another acute diabetic complication.

Contraindications

- Do not give oral glucose, food, drink, or oral medication to a patient who cannot swallow reliably or protect the airway.
- Do not administer insulin or independently alter an insulin-pump dose under this protocol.
- Do not administer routine IV fluid solely because the glucose is elevated; fluid is reserved for hypotension or poor perfusion.

Assessment | Immediate assessment

- Assess airway protection, ventilation, oxygenation, circulation, mental status, focal neurologic findings, seizure activity, trauma, temperature, and blood glucose.
- If a continuous glucose monitor value is inconsistent with the clinical presentation, confirm with the agency glucometer when feasible without delaying treatment of severe suspected hypoglycemia.
- Determine the diabetes type, usual glucose range, insulin and oral medication names, last medication dose, last meal, recent medication changes, pump or continuous-monitor use, and treatment given before EMS arrival.
- Ask about intentional overdose, alcohol use, prolonged fasting, vomiting, diarrhea, infection symptoms, pregnancy, renal or liver disease, and previous recurrent hypoglycemia.

Assessment | Hyperglycemic emergency assessment

- A glucose at least 250 mg/dL should prompt evaluation for dehydration, Kussmaul respirations, abdominal pain, nausea or vomiting, infection, altered mental status, hypotension, poor perfusion, DKA, and HHS.
- Consider euglycemic DKA even when glucose is below 250 mg/dL, especially with SGLT2-inhibitor use, pregnancy, prolonged fasting, vomiting, alcohol use, or significant acute illness.
- Inspect insulin-pump tubing and the insertion site without manipulating or disconnecting the system. Ask about pump alarms, interruption, empty reservoir, dislodgement, or recent dose changes.

Treatment steps

- Treat glucose below 70 mg/dL when symptoms are present or when symptoms cannot be assessed reliably because of altered mental status.
- If swallowing and airway protection are intact, give oral glucose 15-25 g according to the packaged dose. Reassess clinically and repeat glucose within 10-15 minutes.
- If oral treatment is unsafe or ineffective, give D10 up to 250 mL IV/IO, titrated to improving mental status and glucose at least 70 mg/dL, maximum 25 g. Recheck glucose and neurologic status after five minutes.
- If vascular access cannot be obtained promptly, give glucagon 1 mg IM. Position for aspiration protection and continue airway monitoring.
- Repeat or continue titrated D10 within the 25 g maximum when hypoglycemia and symptoms persist. Do not create unnecessary hyperglycemia after the treatment target is reached.
- For persistent altered mental status or focal neurologic findings after glucose correction, continue evaluation and treatment under the altered-mental-status, stroke, seizure, overdose, trauma, or sepsis pathway as appropriate.
- For hyperglycemia, provide supportive care and treat hypotension or poor perfusion with isotonic crystalloid. Do not administer insulin or routine maintenance fluid.
- A capable patient may stop or suspend their own insulin pump. EMS must not independently change the programmed dose; transport suspected pump failure or DKA.
- Strongly recommend transport for oral hypoglycemic or sulfonylurea use, long-acting insulin, recurrent hypoglycemia, seizure, injury, intentional overdose, pregnancy, infection, renal or liver disease, uncertain cause, persistent neurologic abnormality, suspected DKA/HHS, or insulin-pump failure.

Medications

Oral glucose

15-25 g PO according to the packaged dose

- EMT, AEMT, and Paramedic when glucose is below 70 mg/dL with symptoms and the patient is awake, follows commands, swallows reliably, and protects the airway.
- Rapid-carbohydrate food or drink is acceptable when commercial oral glucose is unavailable and swallowing is safe.
- Reassess clinically and repeat glucose within 10-15 minutes.

Dextrose 10% (D10)

Up to 250 mL IV/IO, titrated to improving mental status and glucose ≥70 mg/dL; maximum 25 g

- AEMT and Paramedic when oral glucose is unsafe or ineffective.
- Recheck glucose and neurologic status after five minutes.
- Stop titration when the clinical and glucose targets are reached.

Glucagon

1 mg IM

- AEMT and Paramedic when oral glucose is unsafe and vascular access cannot be obtained promptly.
- Protect against aspiration because vomiting may occur.
- May be ineffective in malnutrition, prolonged fasting, chronic alcohol use, severe liver disease, or depleted glycogen states.

Isotonic crystalloid

500 mL IV/IO aliquots; repeat with reassessment; maximum 2 L

- For hypotension or poor perfusion only - not an elevated glucose value alone.
- Use 250 mL aliquots for CHF, renal failure, dialysis, liver failure, pulmonary edema, or other volume-overload risk.
- Do not begin a routine maintenance infusion under AM-02.

Warnings / precautions

- Do not administer anything orally when swallowing or airway protection is impaired.
- Glucagon may be ineffective when hepatic glycogen is depleted and may cause vomiting.
- Sulfonylureas, other oral hypoglycemic agents, and long-acting insulin may cause delayed or recurrent hypoglycemia after an initially normal repeat glucose.
- A normal glucose after treatment does not explain persistent altered mental status, focal deficit, seizure, or shock; continue evaluation for another emergency.
- DKA can occur below 250 mg/dL, particularly with SGLT2 inhibitors, pregnancy, fasting, vomiting, alcohol use, or acute illness.
- Do not administer insulin, change an insulin-pump setting, or give routine fluid solely for hyperglycemia.

Clinical pearls

- Persistent focal neurologic deficit after glucose correction remains a stroke alert until proven otherwise.
- D10 permits controlled titration and reduces the hypertonic exposure associated with concentrated D50 administration.
- Quality-control checks must be maintained for agency glucometers according to manufacturer and agency policy.
- For possible oral-agent overdose or uncertain medication effect, contact Poison Control early and provide the exact product, strength, amount, and ingestion time when available.
- Document the initial glucose, neurologic status, treatment dose and route, response, repeat glucose values, meal availability, medication risks, capacity assessment, and transport recommendation.

Special populations | Refusal after hypoglycemia

- Refusal may be considered only when the patient has known diabetes and a clear correctable cause; normal mental status, neurologic examination, and decision-making capacity; glucose at least 80 mg/dL and stable on repeat; ability to eat a meal containing longer-acting carbohydrate; and a responsible adult remaining with access to further glucose monitoring.
- Transport is strongly recommended and refusal is high risk when oral hypoglycemic or sulfonylurea use, long-acting insulin, overdose, recurrence, seizure, injury, pregnancy, infection, renal or liver disease, uncertain cause, persistent deficit, lack of food, or absence of a responsible adult is present.
- Contact Medical Control whenever any refusal criterion or recurrence risk is uncertain.

Special populations | Pregnancy

- Treat maternal hypoglycemia promptly using the same adult dosing.
- Consider euglycemic DKA with vomiting, illness, fasting, SGLT2 exposure, or compatible symptoms even when glucose is below 250 mg/dL.

Special populations | Renal, hepatic, or cardiac disease

- Use 250 mL fluid aliquots for hypotension or poor perfusion and reassess lungs, oxygenation, blood pressure, and perfusion after each aliquot.
- Recognize increased recurrence risk from impaired medication clearance and possible poor glucagon response with severe liver disease.

Related pathways / contacts

Call Poison Control - National Poison Control Center: 1-800-222-1222.

UP-04 Altered Mental Status - Open for persistent altered mental status or another suspected metabolic, toxicologic, or neurologic cause.

UP-13 Seizure - Open for active seizure or persistent postictal findings.

UP-14 Stroke - Open when focal neurologic deficits persist after glucose correction.

AM-05 Hypotension / Shock - Open for persistent hypotension or poor perfusion.

Source and references

American Diabetes Association Professional Practice Committee. Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises: Standards of Care in Diabetes - 2026. | American Diabetes Association Professional Practice Committee. Pharmacologic Approaches to Glycemic Treatment: Standards of Care in Diabetes - 2026. | National Association of State EMS Officials. National Model EMS Clinical Guidelines, Version 3, 2022. | North Carolina College of Emergency Physicians. AM-02 Diabetic - Adult Protocol, revised September 1, 2025. | Claiborne County EMS formulary and medical-director decisions, October 2026.

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